Physics 9th – Chapter-1-Physical Quantities and Measurement (Short Response Questions)

Q1. How physics plays an important role in our life?

Ans:

- Physics is the most fundamental science; it studies matter, energy and their interaction.
- Its principles are the basis of modern technology: electricity, mobile phones, computers, internet, X-rays, MRI and satellites.
- Other sciences also use physics, e.g. biology uses fluid movement to explain blood flow in the heart and veins.
- Concept: physics makes our life easy, comfortable and safe.

Q2. Estimate your age in minutes and seconds.

Ans:

- Suppose age = 15 years; 1 year = 365 days, 1 day = 24 h, 1 h = 60 min, 1 min = 60 s.
- Age in minutes = $15 \times 365 \times 24 \times 60 = 7\,884\,000$ min $\approx 7.88 \times 10^6$ min.
- Age in seconds = $7\,884\,000 \times 60 = 473\,040\,000$ s $\approx 4.73 \times 10^8$ s.
- Concept: units are converted by multiplying with conversion factors (ignoring leap years).

Q3. What base quantities are involved in these derived physical quantities: force, pressure, power and charge?

Ans:

- Force ($F = ma$), unit N = kg m s^{-2}: mass, length and time.
- Pressure ($P = F/A$), unit Pa = kg m^{-1} s^{-2}: mass, length and time.
- Power ($P = W/t$), unit W = kg m^2 s^{-3}: mass, length and time.
- Charge ($Q = It$), unit C = A s: electric current and time.

Q4. Show that prefix micro is thousand times smaller than prefix milli.

Ans:

- Milli (m) = 10^{-3} and micro (μ) = 10^{-6}.
- Ratio: milli / micro = $10^{-3} / 10^{-6} = 10^3 = 1000$.
- So 1 mm = 1000 μm, i.e. micro is one thousand times smaller than milli.
- Concept: prefixes are multiples or sub-multiples of units in powers of ten.

Q5. Justify that displacement is a vector quantity while energy is a scalar quantity.

Ans:

- Displacement is the shortest distance between two points in a particular direction, e.g. 5 m towards east.
- It needs both magnitude and direction, so it is a vector.
- Energy (e.g. 100 J) is completely described by magnitude with unit; it has no direction, so it is a scalar.
- Concept: vectors follow vector addition; scalars are added by ordinary algebra.

Q6. Screw gauge gives more precise length than vernier callipers. Briefly explain why?

Ans:

- Least count of vernier callipers is 0.1 mm (0.01 cm).
- Least count of screw gauge is 0.01 mm (0.001 cm), which is ten times smaller.
- An instrument with smaller least count measures smaller changes, so the screw gauge is more precise.
- Concept: precision of a measurement depends on the least count of the instrument.

Q7. Differentiate between mechanical stopwatch and digital stopwatch.

Ans:

- Mechanical (analogue) stopwatch works by winding a spring; it has a dial with a needle; least count is about 1 s (0.1 s in some).
- Digital stopwatch works on a battery; it shows time on a digital display; least count is 0.1 s or even 0.01 s.
- Digital stopwatch is more precise and easier to read than a mechanical one.
- Concept: both measure time intervals; smaller least count gives more precise time.

Q8. How is a measuring cylinder used to measure the volume of an irregular shaped stone?

Ans:

- Pour some water in the measuring cylinder and note the initial volume V_1.
- Tie the stone with a thread and lower it gently so that it is completely immersed; note the new level V_2.
- Volume of the stone = $V_2 - V_1$ (the water displaced by the stone).
- Concept: an immersed object displaces water equal to its own volume.

Q9. What precautions should be kept in mind while taking measurement using a measuring cylinder?

Ans:

- Place the cylinder vertically on a flat horizontal surface.
- Keep the eye level with the bottom of the meniscus (top of the meniscus for mercury) to avoid parallax error.
- Use a cylinder of suitable size and least count, and read after the liquid becomes still.
- Concept: these precautions reduce personal (systematic) errors.

Q10. Why do we need to consider significant digits in measurements?

Ans:

- Significant figures are all the accurately known digits plus the first doubtful digit.
- They show the precision of a measurement and the least count of the instrument used.
- They prevent writing more (false) or fewer digits than are actually measured, also in calculated results.
- Concept: more significant figures mean a more precise measurement.

Q11. How random error can be reduced?

Ans:

- Random errors occur due to unknown, unpredictable changes, e.g. reaction time or sudden change in conditions.
- They are reduced by taking a number of readings and finding their average (mean) value.
- E.g. measure the time of 10 or 20 oscillations of a pendulum and divide by their number.
- Concept: random errors cannot be removed completely, only reduced.

Q12. Differentiate between precision and accuracy.

Ans:

- Precision: how close repeated measurements of the same quantity are to each other.
- Accuracy: how close a measured value is to the true or accepted value.
- Precision depends on the least count of the instrument; accuracy depends on removing errors (e.g. zero error).
- Concept: a measurement can be precise but not accurate, e.g. a thermometer that always reads 2 °C too high.